COMP 4910 Senior Design Project I, Fall 2026 PAF: Project Assignment Form, v4.0	
Project Codename	**HORUS**
Project Title	**HORUS: A Latency-Tolerant, Safety-First Drone Directed by a Cloud AI Commander for GPS-Denied Search and Rescue**
Project Team: • **Student 1:** • **Student 2:** • **Student 3:** • **Student 4:**	For Each Student: ID Number, Name and Lastname, E-Mail Address, Mobile Phone Number and Department (if not COMP) 1. 2. 3. 4.
Project: Brief Summary Write some details of your problem definition here (this is your RSD v0.0). Writing a single sentence that is ambiguous is not acceptable but do not write multiple pages either. One or two paragraphs acceptable. Remember that the size of the **abstract** of a scientific paper is a good indication." "	Drones today are either flown by a skilled pilot or run a fixed task such as "follow me". Frontier vision-language models (Claude, GPT) understand open-ended requests like "find the person in the red shirt", but they are too large to run onboard and answer in 0.5–3 s over a network, while a drone must react within milliseconds. HORUS asks: how can a slow, remote AI reliably and safely direct a fast drone over a delayed or intermittent link? HORUS splits control into three tiers: a cloud commander decides what to do, an onboard executor decides how, and the ArduPilot flight controller decides whether it is safe. The drone tracks candidates with persistent IDs and sends a numbered (set-of-mark) frame; the commander replies with an ID and a bounded skill such as follow or approach via MCP tools, so a late answer stays valid. The executor compensates link delay, clamps speed, distance and geofence, and holds or returns home when commands expire. The target is GPS-free indoor search and rescue. Deliverables: onboard software, a web operator console, a SITL + Gazebo test bench and a real quadcopter, evaluated by task success and zero AI-caused collisions under 0–3 s added latency against a cloud-only baseline.
Project Advisor: Name and Lastname Signature and Date	Assoc. Prof. Dr. Arman SAVRAN
COMP 4910 Coordinator: **Signature and Date**	Ahmet Hasan KOLTUKSUZ

Procedure for COMP 4910 Project Assignment

1. Read project descriptions posted by COMP project advisors, discuss projects you are interested in with your fellow students, project advisors and decide on a project. You can also propose your own project to any of the COMP project advisors.
2. Project team size is 4-to-5 team members/students. 5 is difficult, more than 5 is almost impossible. One-student or two-student projects will not be allowed, except only in very exceptional cases.
3. Teamwork projects, where some team members are from other departments of Yaşar University or from industry, are encouraged.
4. Fill the Project Assignment Form (PAF) out and get the approval of your project advisor.
5. Get the approval of COMP 4910 Coordinator.
6. Give a copy of PAF, approved by COMP 4910 Coordinator, to your project advisor and to the Coordinator, either in paper form or by emailing a scanned copy. Then, your COMP 4910 project assignment is complete.
7. Repeat above steps if you want have any modification in your project assignment.
8. Note that you will continue your COMP 4910 project as COMP 4920 in the following semester.